Manual of data.wl: Conventions and Notations

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The matching results including Higher-order h-polynomials are provided in the Mathematica data file `data.wl`. The file can be loaded in Mathematica with

```
data = Get["path/to/data.wl"]
```

and the results can be accessed from the `data` using the nested keys. For example, the coefficient labeled C_{V^2} in the pHEFT can be obtained by

```
data[Leff][pHEFT][CV2]
```

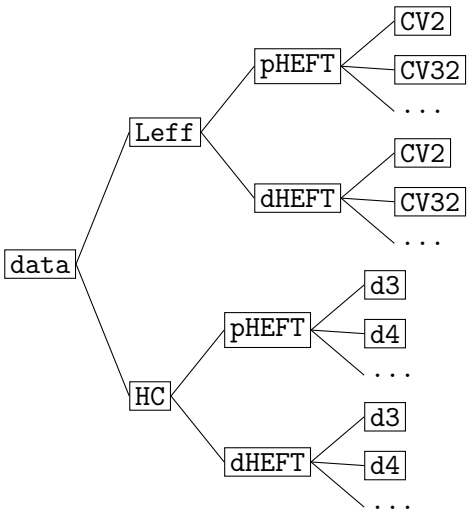

Figure 1. Tree structure of `data.wl`. The file contains two main categories, `Leff` and `HC`, each further divided into `pHEFT` and `dHEFT`, containing the corresponding Wison coefficients or HEFT couplings.

The structure of the data file is shown in Fig. 1. It contains two main categories: `Leff` and `HC`. The former includes the Wilson coefficients associated with the operators defined in Tables 1, 2, and 3, while the latter contains the HEFT couplings used directly for phenomenological analyses (e.g., numerical plots). Each category is further divided into two sub-keys, `pHEFT` and `dHEFT`, containing the corresponding Wilson coefficients or HEFT couplings.

The innermost Wilson coefficient entries and HEFT coupling entries include an auxiliary parameter t labeling the power-counting order. For instance, the $\mathcal{O}(t^0)$ part of C_{V^2} in pHEFT can be extracted via

```
Coefficient[data[Leff][pHEFT][CV2], t, 0]
```

while the full expression can be obtained by setting $t \to 1$:

```
data[Leff][pHEFT][CV2] /. t -> 1
```

Here `/.` denotes the standard `ReplaceAll` operator in Mathematica.

1 Wilson Coefficients of Operators

Tables 1, 2, and 3 list all operators appearing in the matching results, together with their Wilson coefficient symbols and Mathematica key names: bosonic operators with zero or two derivatives (Table 1), bosonic operators with four derivatives (Table 2), and operators involving fermions (Table 3).

Operators	WC symbols	Key names
$\mathbb{1}$	$V(h)$	`Vh`
$\langle V_\mu V^\mu \rangle$	C_{V^2}	`CV2`
$\langle V_\mu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle$	$C_{V_3^2}$	`CV32`

Table 1. p^0 and p^2 operators with Wilson coefficient symbols and Mathematica key names. Both $V(h)$ and C are polynomials in h.

Operators	WC symbols	Key names
$\langle V_\mu V^\mu \rangle \langle V_\nu V^\nu \rangle$	$C_{V^2V^2}$	`CV2V2`
$\langle V_\mu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle \langle V_\nu V^\nu \rangle$	$C_{V_3^2V^2}$	`CV32V2`
$\langle V_\mu \sigma_3 \rangle \langle V_\nu \sigma_3 \rangle \langle V^\mu V^\nu \rangle$	$C_{V_3V_3VV}$	`CV3V3VV`
$\langle V_\mu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle \langle V_\nu \sigma_3 \rangle \langle V^\nu \sigma_3 \rangle$	$C_{V_3^2V_3^2}$	`CV32V32`
$\langle V_\mu V_\nu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle D^\nu h$	$C_{VV_3V_3Dh}$	`CVV3V3Dh`
$\left[2 \langle V_\mu V_\nu \rangle - \langle V_\mu \sigma_3 \rangle \langle V_\nu \sigma_3 \rangle\right] D^\mu h D^\nu h$	C_{VVDhDh}	`CVVDhDh`
$\langle V_\mu V^\mu \rangle D_\nu h D^\nu h$	$C_{V^2Dh^2}$	`CV2Dh2`
$\langle V_\mu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle D_\nu h D^\nu h$	$C_{V_3^2Dh^2}$	`CV32Dh2`
$D_\mu h D^\mu h D_\nu h D^\nu h$	$C_{Dh^2Dh^2}$	`CDh2Dh2`

Table 2. p^4 operators with Wilson coefficient symbols and Mathematica key names. Each C is a polynomial in h.

Operators	WC symbols	Key names
$\bar{Q}_L U \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix} Q_R \langle V_\mu V^\mu \rangle + \text{h.c.}$	$C_{Q^2V^2}$	`CQ2V2`
$\bar{Q}_L U \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix} Q_R \langle V_\mu \sigma_3 \rangle \langle V^\mu \sigma_3 \rangle + \text{h.c.}$	$C_{Q^2V_3^2}$	`CQ2V32`
$\bar{Q}_L U \begin{pmatrix} 0 & -y_d \\ y_u & 0 \end{pmatrix} Q_R \langle V_\mu \sigma_1 \rangle \langle V^\mu \sigma_3 \rangle$ $+ \bar{Q}_L U \begin{pmatrix} 0 & \mathrm{i}y_d \\ \mathrm{i}y_u & 0 \end{pmatrix} Q_R \langle V_\mu \sigma_2 \rangle \langle V^\mu \sigma_3 \rangle + \text{h.c.}$	$C_{Q^2V_{1,2}V_3}$	`CQ2V12V3`
$\bar{Q}_L U \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix} Q_R D_\mu h D^\mu h + \text{h.c.}$	$C_{Q^2Dh^2}$	`CQ2Dh2`
$\bar{Q}_L U \begin{pmatrix} 0 & y_d \\ y_u & 0 \end{pmatrix} Q_R \langle V_\mu \sigma_1 \rangle D^\mu h$ $+ \bar{Q}_L U \begin{pmatrix} 0 & -\mathrm{i}y_d \\ \mathrm{i}y_u & 0 \end{pmatrix} Q_R \langle V_\mu \sigma_2 \rangle D^\mu h + \text{h.c.}$	$C_{Q^2V_{1,2}Dh}$	`CQ2V12Dh`
$\bar{Q}_L U \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix} Q_R + \text{h.c.}$	C_{Q^2}	`CQ2`
$\left[\bar{Q}_L U \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix} Q_R + \text{h.c.} \right]^2$	$C^1_{Q^4}$	`CQ41`
$\left[\bar{Q}_L U \begin{pmatrix} 0 & y_d \\ 0 & 0 \end{pmatrix} Q_R - \bar{Q}_R \begin{pmatrix} 0 & y_u \\ 0 & 0 \end{pmatrix} U^\dagger Q_L \right] \times \text{h.c.}$	$C^2_{Q^4}$	`CQ42`

Table 3. Fermionic operators with Wilson coefficient symbols and Mathematica key names. Each C is a polynomial in h.

Each operator in Table 3 contains a constant matrix that can be decomposed into the standard $\{I, \sigma_i\}$ basis. To facilitate comparison with existing literature, we map the operators from Table 3 onto those in Table 4. For the four-fermion operators, Fierz rearrangements are employed to achieve the final form.

Operators	WCs
$(\bar{Q}_L U Q_R)\langle V_\mu V^\mu\rangle + \text{h.c.}$	$\frac{1}{2}(y_u + y_d)C_{Q^2V^2}$
$(\bar{Q}_L U \sigma_3 Q_R)\langle V_\mu V^\mu\rangle + \text{h.c.}$	$\frac{1}{2}(y_u - y_d)C_{Q^2V^2}$
$(\bar{Q}_L U Q_R)\langle V_\mu \sigma_3\rangle\langle V^\mu \sigma_3\rangle + \text{h.c.}$	$\frac{1}{2}(y_u + y_d)C_{Q^2V_3^2}$
$(\bar{Q}_L U \sigma_3 Q_R)\langle V_\mu \sigma_3\rangle\langle V^\mu \sigma_3\rangle + \text{h.c.}$	$\frac{1}{2}(y_u - y_d)C_{Q^2V_3^2} - \frac{1}{2}(y_u - y_d)C_{Q^2V_{1,2}V_3}$
$(\bar{Q}_L U [V_\mu, \sigma_3] Q_R)\langle V^\mu \sigma_3\rangle + \text{h.c.}$	$\frac{1}{2}(y_u + y_d)C_{Q^2V_{1,2}V_3}$
$(\bar{Q}_L U V_\mu Q_R)\langle V^\mu \sigma_3\rangle + \text{h.c.}$	$(y_u - y_d)C_{Q^2V_{1,2}V_3}$
$(\bar{Q}_L U Q_R) D_\mu h D^\mu h + \text{h.c.}$	$\frac{1}{2}(y_u + y_d)C_{Q^2Dh^2}$
$(\bar{Q}_L U \sigma_3 Q_R) D_\mu h D^\mu h + \text{h.c.}$	$\frac{1}{2}(y_u - y_d)C_{Q^2Dh^2}$
$(\bar{Q}_L U [V_\mu, \sigma_3] Q_R) D^\mu h + \text{h.c.}$	$\frac{1}{2}(y_u - y_d)C_{Q^2V_{1,2}Dh}$
$(\bar{Q}_L U V_\mu Q_R) D^\mu h + \text{h.c.}$	$(y_u + y_d)C_{Q^2V_{1,2}Dh}$
$(\bar{Q}_L U \sigma_3 Q_R)\langle V_\mu \sigma_3\rangle D^\mu h + \text{h.c.}$	$-\frac{1}{2}(y_u + y_d)C_{Q^2V_{1,2}Dh}$
$(\bar{Q}_L U Q_R) + \text{h.c.}$	$\frac{1}{2}(y_u + y_d)C_{Q^2}$
$(\bar{Q}_L U \sigma_3 Q_R) + \text{h.c.}$	$\frac{1}{2}(y_u - y_d)C_{Q^2}$
$(\bar{Q}_L U Q_R)(\bar{Q}_L U Q_R) + \text{h.c.}$	$\frac{1}{4}(y_u + y_d)^2 C^1_{Q^4}$
$(\bar{Q}_L U Q_R)(\bar{Q}_L U \sigma_3 Q_R) + \text{h.c.}$	$\frac{1}{2}(y_u^2 - y_d^2)C^1_{Q^4}$
$(\bar{Q}_L U \sigma_3 Q_R)(\bar{Q}_L U \sigma_3 Q_R) + \text{h.c.}$	$\frac{1}{4}(y_u - y_d)^2 C^1_{Q^4} + \frac{1}{4}y_u y_d C^2_{Q^4}$
$(\bar{Q}_L \sigma^I U Q_R)(\bar{Q}_L \sigma^I U Q_R) + \text{h.c.}$	$-\frac{1}{4}y_u y_d C^2_{Q^4}$
$(\bar{Q}_L \gamma_\mu Q_L)(\bar{Q}_R \gamma^\mu Q_R)$	$\frac{1}{4}(y_u^2 + y_d^2)C^1_{Q^4} + \frac{1}{8}(y_u^2 + y_d^2)C^2_{Q^4}$
$(\bar{Q}_L \gamma_\mu \sigma^I Q_L)(\bar{Q}_R \gamma^\mu U^\dagger \sigma^I U Q_R)$	$\frac{1}{2}y_u y_d C^1_{Q^4}$
$(\bar{Q}_L \gamma_\mu U \sigma_3 U^\dagger Q_L)(\bar{Q}_R \gamma^\mu Q_R)$	$\frac{1}{4}(y_u^2 - y_d^2)C^1_{Q^4} - \frac{1}{8}(y_u^2 - y_d^2)C^2_{Q^4}$
$(\bar{Q}_L \gamma_\mu Q_L)(\bar{Q}_R \gamma^\mu \sigma_3 Q_R)$	$\frac{1}{4}(y_u^2 - y_d^2)C^1_{Q^4} + \frac{1}{8}(y_u^2 - y_d^2)C^2_{Q^4}$
$(\bar{Q}_L \gamma_\mu U \sigma_3 U^\dagger Q_L)(\bar{Q}_R \gamma^\mu \sigma_3 Q_R)$	$\frac{1}{4}(y_u - y_d)^2 C^1_{Q^4} - \frac{1}{8}(y_u^2 + y_d^2)C^2_{Q^4}$

Table 4. The mapped fermionic operators and their Wilson coefficients after basis transformation and Fierz rearrangement.

2 HEFT Couplings

The HEFT couplings used for the numerical analysis are summarized in Tables 5 and 6. Specifically, the couplings in Table 6 are extracted from the constant components of the Wilson coefficients associated with the p^4 operators in the Lagrangian.

HCs	Key names
$\Delta\kappa_3$	`d3`
$\Delta\kappa_4$	`d4`
Δa	`a`
Δb	`b`
$\Delta\alpha$	`c1`
$\Delta a^{\not{c}}$	`c2`
$\Delta b^{\not{c}}$	`c3`

Table 5. Leading-order HEFT couplings with Mathematica key names.

Operators	Key names for HC
$\langle V_\mu V^\mu\rangle \langle V_\nu V^\nu\rangle$	`p41`
$\langle V_\mu \sigma_3\rangle \langle V^\mu \sigma_3\rangle \langle V_\nu V^\nu\rangle$	`p42`
$\langle V_\mu \sigma_3\rangle \langle V_\nu \sigma_3\rangle \langle V^\mu V^\nu\rangle$	`p43`
$\langle V_\mu \sigma_3\rangle \langle V^\mu \sigma_3\rangle \langle V_\nu \sigma_3\rangle \langle V^\nu \sigma_3\rangle$	`p44`
$2\langle V_\mu V_\nu \sigma_3\rangle \langle V^\mu \sigma_3\rangle D^\nu h$	`p45`
$\left[2\langle V_\mu V_\nu\rangle - \langle V_\mu \sigma_3\rangle \langle V_\nu \sigma_3\rangle\right] D^\mu h D^\nu h$	`p46`
$\langle V_\mu V^\mu\rangle D_\nu h D^\nu h$	`p47`
$\langle V_\mu \sigma_3\rangle \langle V^\mu \sigma_3\rangle D_\nu h D^\nu h$	`p48`
$D_\mu h D^\mu h D_\nu h D^\nu h$	`p49`

Table 6. p^4 operators and the Mathematica key names for the corresponding HEFT couplings.